

Diabetes Mellitus



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Outline

- Pathophysiology
- Diagnosis criteria
- Complications
- Treatment Goals
- Nonpharmacologic Therapy
- Pharmacologic Therapy



Abbreviations

- **ADA:** American Diabetes Association
- **DPP4i:** dipeptidyl peptidase 4 inhibitor
- **GLP1:** glucagon-like peptide 1 agonists
- **NPO:** *nil per os* (nothing by mouth)
- **SGLT2i:** Sodium glucose co-transporter 2 inhibitor
- **SJS:** Steven-Johns Syndrome
- **SU:** sulfonylurea
- **TZD:** thiazolidinediones
- **UTI:** urinary tract infection



Objectives

1. Differentiate pathophysiology between type 1 and type 2 diabetes
2. Understand value of A1c testing in diagnosis and management of diabetes
3. Identify common complications associated with diabetes
4. Recognize basic mechanism of action for medications used in the treatment of diabetes



Pathophysiology

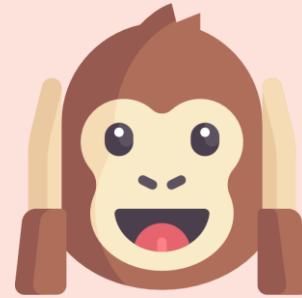
Type 1



Pancreatic beta-cell
destruction leads to
absolute insulin deficiency



Type 2



Insulin resistance and
progressive pancreatic beta-
cell dysfunction

American Diabetes Association Diagnostic Criteria

Hemoglobin A1C $\geq$ **6.5%**

Classic **symptoms** of hyperglycemia: polyuria, polydipsia, polyphagia, unexplained weight loss, fatigue



Hyperglycemic crisis AND random glucose $\geq$ **200 mg/dL**

Fasting plasma glucose $\geq$ **126 mg/dL**

Fasting defined as no intake for at least 8 hours

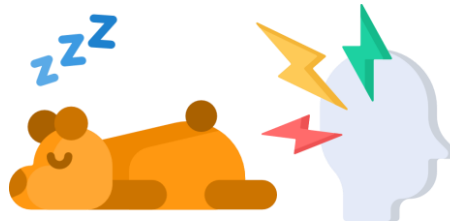
2-hour plasma glucose $\geq$ **200 mg/dL** after a 75 g oral glucose tolerance test

Acute Complications: Hypoglycemia

Plasma glucose < 70 mg/dL

Symptoms

- Mild: tremor, dizziness, palpitations, sweating, hunger, irritability
- Severe: loss of consciousness, seizures, coma



Acute Complications: Hypoglycemia



Can **mask** hypoglycemia:
Beta-blockers



Can **cause** hypoglycemia:
Sulfonylureas, insulin, metglinides,
alpha-glucosidase inhibitors,
pramlintide

Acute Complications: Hypoglycemia (continued)

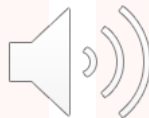
Treatment: Eat **15 grams** carbohydrates. Wait **15 minutes**, then check blood glucose. Repeat if blood glucose still < 80 mg/dL. Follow with meal.



4 oz fruit juice



4 oz regular soda



8 oz low-fat milk

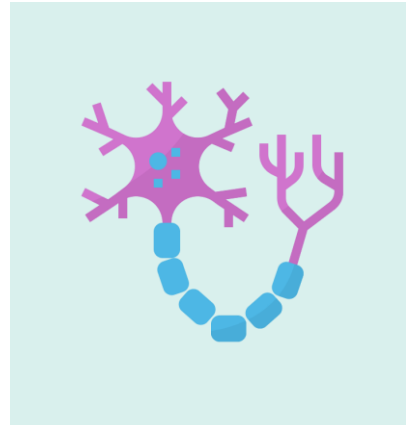
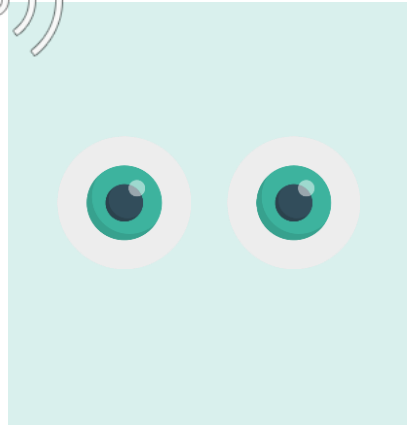
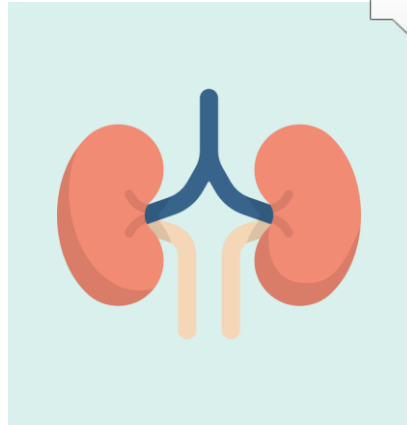


3 teaspoons
granulated sugar

Chronic Complications

Macrovascular: coronary heart disease, stroke, peripheral arterial disease

Microvascular: nephropathy, retinopathy, neuropathy



Treatment Goals

Hemoglobin **A1C** < 7%

Pre-prandial plasma glucose: **80-130 mg/dL**

Peak **post**-prandial plasma glucose: < **180 mg/dL**



Type 2 Diabetes Treatment

First-line

Lifestyle
modifications



Monotherapy

Add
Metformin



Dual Therapy

Add **second** agent:



- SGLT2 inhibitor
- GLP1 agonist
- SU
- TZD
- DPP4 inhibitor
- Insulin

Triple Therapy

Add **third** agent.

Not recommended:

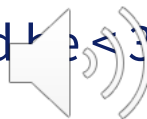
- Metformin + DPP4 inhibitor + GLP1 agonist
- Metformin + SU + insulin

Lifestyle Modifications



Weight loss

- Encouraged to lose > 5% of body weight in 3 months
 - Reducing weekly calorie intake by 3500 kcal = 1 lbs weight loss
 - Waist circumference should be < 35 inches for females, < 40 inches for males
- Patients should get up and move every 30 minutes
 - 150 minutes of moderate intensity aerobic activity/week



Patients should be encouraged to stop smoking

Pharmacologic Therapy

Oral therapy:

- Biguanide
- Sulfonylureas
- Meglitinides
- Thiazolidinediones
- SGLT2i
- DPP4
- Alpha-Glucosidase inhibitors



Injectable Therapy:

- GLP1 agonists
- Amylinomimetics
- Insulin

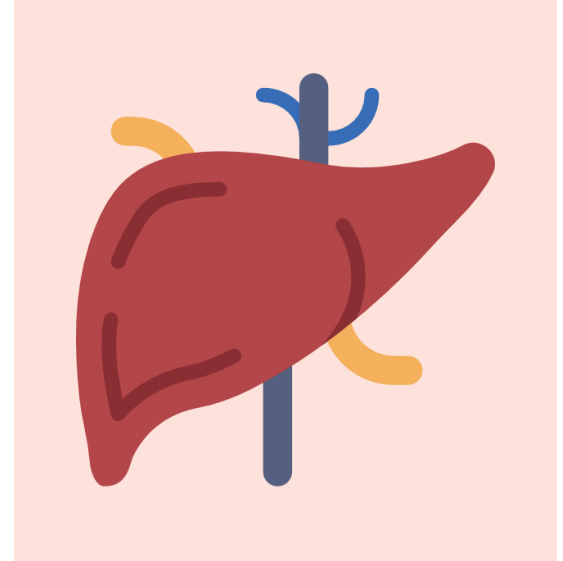


Oral Therapy

Biguanides

Metformin (*Glucophage*)

- Mechanism of action: decreases hepatic glucose output, increases insulin sensitivity
- Side effects: abdominal cramping, nausea, vomiting, diarrhea
- Dosing: once to twice daily



Sulfonylureas

Glipizide (*Glucotrol*), **Glimepiride** (*Amaryl*), **Glyburide** (*Glynase*)

- Mechanism of action: Increases insulin secretion
- Side effects: **hypoglycemia**, weight gain, nausea
- Dosing: **immediate release glipizide** is taken **30 minutes prior** to the first meal of the day, all the rest can be take with the first meal of the day. **Consider holding if NPO.**



Metglinides

Repaglinide (*Prandin*), Nateglinide (*Starlix*)

- Mechanism of action: Increases insulin secretion
- Side effects: **hypoglycemia**, weight gain, headache, upper respiratory tract infection
- Dosing: Up-titrate to three times daily **with meals**



Thiazolidinediones

Pioglitazone (*Actos*), **Rosiglitazone** (*Avandia*)

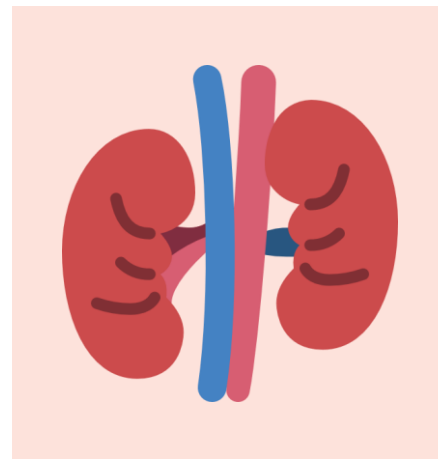
- Mechanism of action: increases insulin sensitivity
- Warnings: peripheral edema
- Side effects: weight gain, upper respiratory tract infections, myalgia
- Dosing: once daily, with or without food



Sodium-Glucose Co-transporter 2 Inhibitors

Canagliflozin (*Invokana*), Dapagliflozin (*Farxiga*), Empagliflozin (*Jardiance*), Ertugliflozin (*Steglatro*)

- Mechanism of action: increases glucose excretion through the urine
- Warnings: ketoacidosis, genital mycotic infections (fournier gangrene)
- Side effects: weight loss, increase in urination, increase in thirst
- Dosing: once daily in the morning



Dipeptidyl Peptidase-4 Inhibitors

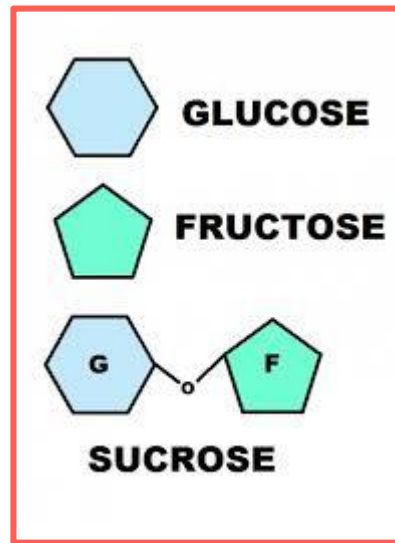
Sita**gliptin** (*Januvia*), Saxa**gliptin** (*Onglyza*), Lina**gliptin** (*Tradjenta*), Alog**gliptin** (*Nesina*)

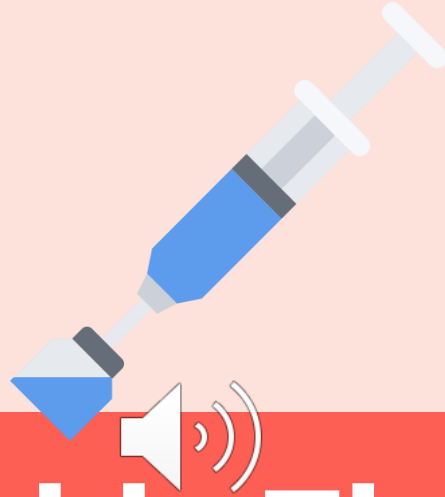
- Mechanism of action: decreases glucagon, which decreases glucose production
- Warnings: acute pancreatitis, severe arthralgia, hypersensitivity reactions (anaphylaxis, SJS) 
- Side effects: upper respiratory tract infections, UTIs, rash - though generally well-tolerated
- Dosing: once daily in the morning

Alpha-glucosidase inhibitors

Acarbose (*Precose*), Miglitol (*Glyset*)

- Mechanism of action: inhibits the metabolism of intestinal sucrose, delays glucose absorption
- Warnings: if hypoglycemia occurs **glucose tablets/gel** must be used
 - **NO** sucrose tablets
- Side effects: flatulence, diarrhea, abdominal pain
- Dosing: Each dose should be taken with the first bite of each meal





Injectable Therapy

Glucagon-like Peptide-1 Agonist

Exenatide (*Byetta*, *Bydureon*), Liraglutide (*Victoza*), Dulaglutide (*Trulicity*), Lixisenatide (*Adlyxin*), Semaglutide (*Ozempic*)

- Mechanism of Action: decreases glucagon, which decreases glucose production; slows gastric emptying, which increases fullness
- Warnings: risk of thyroid C-cell carcinomas, pancreatitis; not recommended in patients with severe GI disease
- Side effects: nausea, weight loss, injection site reactions
- Dosing:
 - Daily: Byetta, Victoza, Adlyxin
 - Weekly: Bydureon, Trulicity, Ozempic

Amylinomimetics

Pramlintide (*SymlinPen*)

- Mechanism of action: decreases glucagon, which decreases glucose production; slows gastric emptying, which increases fullness
- Warnings: severe **hypoglycemia** when used with insulin, contraindicated in patients with gastroparesis
- Side effects: nausea, vomiting, anorexia, headache, weight loss
- Dosing: prior to each major meal (≥ 250 kcal or ≥ 30 g of carbohydrates)
 - **SKIP** if less than this quantity



Insulin

Three main functions:

Required for glucose uptake by muscle cells, which use glucose as energy



Signals liver to make glycogen and prevent its breakdown when glucose is high after a person eats



Regulates fat storage, converting excess glucose to fat and preventing fat breakdown for energy



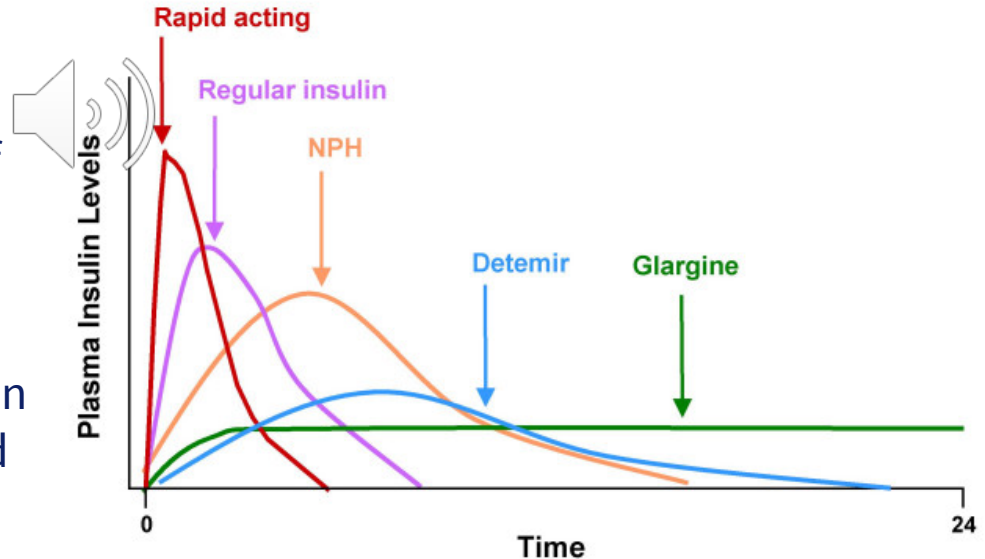
Types of Insulin

Bolus Insulin

- Rapid or short acting insulin
 - Prandial: mealtime insulin to prevent elevations from food
 - Correctional: for acute elevations in response to self monitoring blood glucose

Basal Insulin

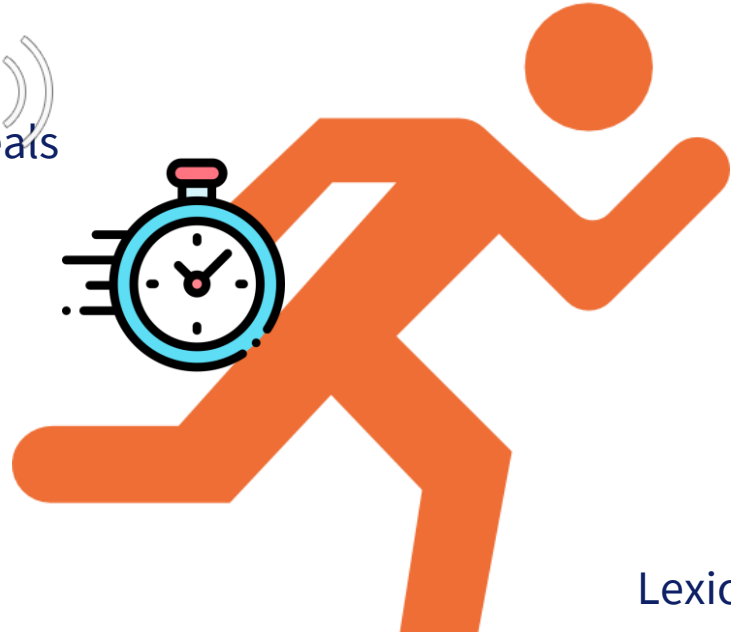
- Long or Intermediate acting insulin
 - Mostly affects FASTING blood glucose



Rapid-Acting Insulin

Aspart (*Novolog, Fiasp*), **Lispro** (*Humalog*)

- Warnings: hypoglycemia, hypokalemia
- Side effects: weight gain, lipodystrophy
- Dosing: give up to 15 minutes before meals



Short-Acting Insulin

Regular Insulin (*Humulin R, Novolin R*)

- Warnings: hypoglycemia, hypokalemia
- Side effects: weight gain, lipodystrophy
- Dosing: give up to 15 minutes before meals



Intermediate-Acting Insulin

NPH insulin (*Humulin N, Novolin N*)

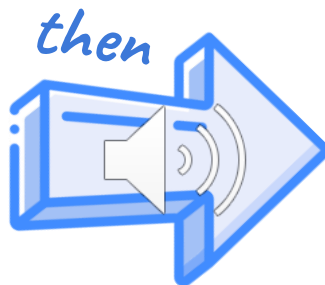
- Warnings: hypoglycemia, hypokalemia
- Side effects: weight gain, lipodystrophy
- Dosing: typically given once or twice daily
- **NPH insulins are cloudy**
 - NPH insulins can be mixed with rapid or short acting insulins



Mixing Two Insulin's



Clear



Cloudy

Long-Acting Insulin

Detemir (*Levemir*), **Glargine** (*Lantus, Toujeo, Basaglar*), **Degludec** (*Tresiba*)

- Warnings: hypoglycemia, hypokalemia
- Side effects: weight gain, lipodystrophy
- Dosing: typically given once or twice daily **at the same time every day**
- **Do not mix with other types of insulins**

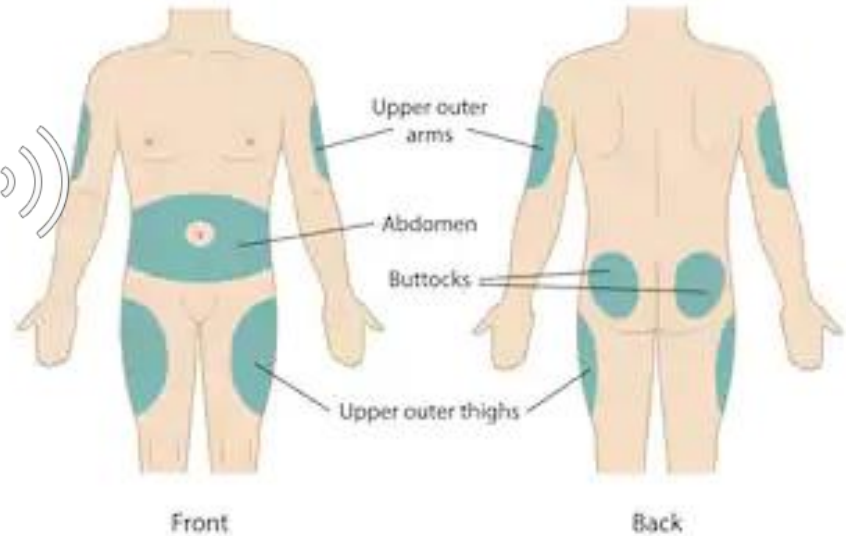


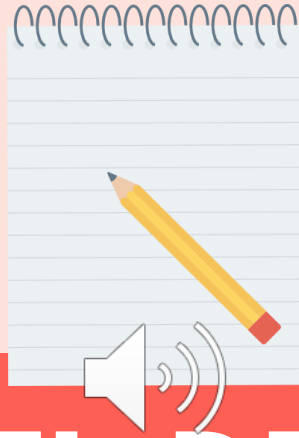
Injection technique

- Rotate injection sites
- Pinch 2 inch portion of the skin and fat between your thumb and first finger and insert the needle all the way at a 90 degree angle
- Press the injection button or plunger all the way down
- Count to 5-10 seconds before removing the needle



Insulin injection sites

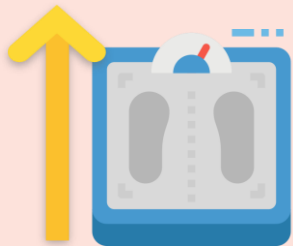




TL;DR

Overall Effect on Weight

Weight **Gain**



Insulin
Sulfonylureas
Metglinides
TZDs



Weight **Loss**



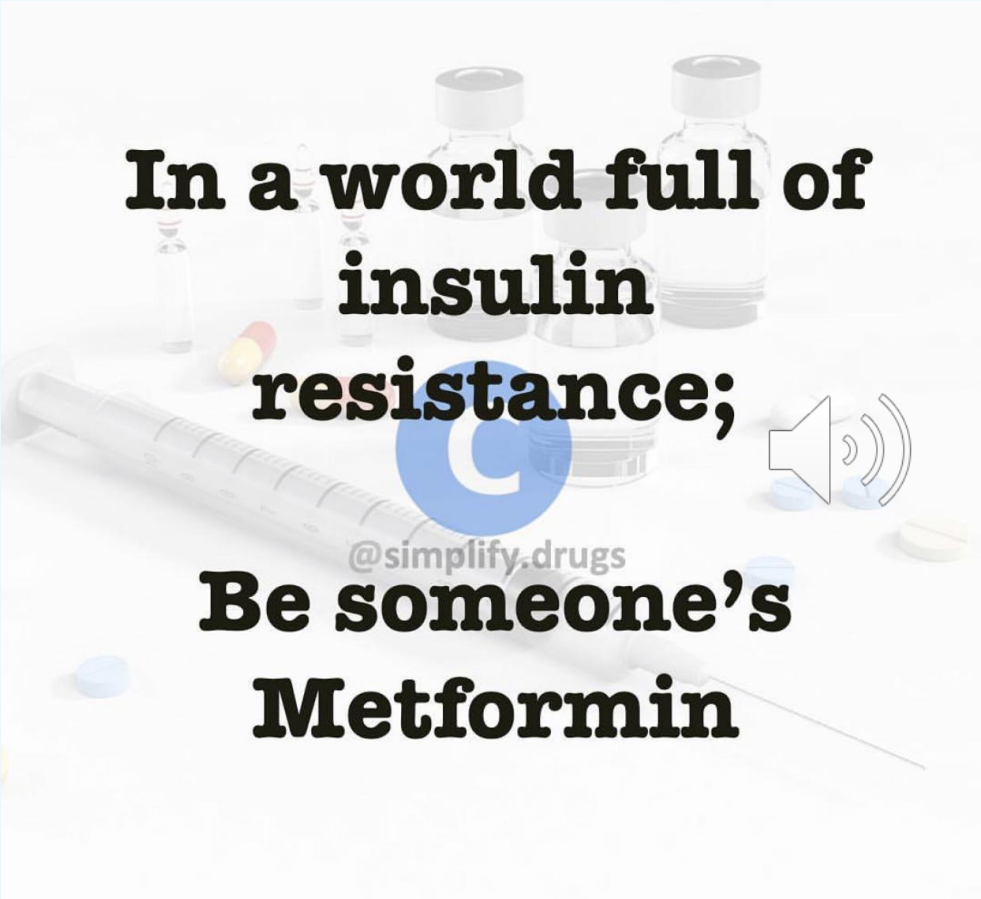
SGLT2 inhibitors
GLP1 agonists
Pramlintide

	MET	GLP1-RA	SGLT2i	DPP4i	AGi	TZD (moderate dose)	SU GLN	COLSVL	BCR-QR	INSULIN	PRAML
HYPO	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Moderate/ Severe Mild	Neutral	Neutral	Moderate to Severe	Neutral
WEIGHT	Slight Loss	Loss	Loss	Neutral	Neutral	Gain	Gain	Neutral	Neutral	Gain	Loss
RENAL / GU	Contra- indicated if eGFR <30 mL/min/ 1.73 m ²	Exenatide Not Indicated CrCl <30 Possible Benefit of Liraglutide	Not Indicated for eGFR <45 mL/ min/1.73 m ² Genital Mycotic Infections Possible CKD Benefit	Dose Adjustment Necessary (Except Linagliptin) Effective in Reducing Albuminuria	Neutral	Neutral	More Hypo Risk	Neutral	Neutral	More Hypo Risk	Neutral
GI Sx	Moderate	Moderate	Neutral	Neutral	Moderate	Neutral	Neutral	Mild	Moderate	Neutral	Moderate
CHF						Moderate	Neutral	Neutral	Neutral	CHF Risk	
CARDIAC ASCVD	Neutral	See #1	See #2	See #3	Neutral	May Reduce Stroke Risk	Possible ASCVD Risk	Benefit	Safe	Neutral	Neutral
BONE	Neutral	Neutral	Neutral	Neutral	Neutral	Moderate Fracture Risk	Neutral	Neutral	Neutral	Neutral	Neutral
KETOACIDOSIS	Neutral	Neutral	DKA Can Occur in Various Stress Settings	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral

- Few adverse events or possible benefits
- Use with caution
- Likelihood of adverse effects

1. Liraglutide—FDA approved for prevention of MACE events.
2. Empagliflozin—FDA approved to reduce CV mortality. Canagliflozin—FDA approved to reduce MACE events.
3. Possible increased hospitalizations for heart failure with alogliptin and saxagliptin.

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A background image showing various medical supplies including insulin vials, a syringe, and pills. A large blue 'G' logo is overlaid in the center.

**In a world full of
insulin
resistance;**



@simplify.drugs
**Be someone's
Metformin**

Thank you!

Questions?

Email us at

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Sources

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